



CSE115: INTRODUCTION TO PROGRAMMING

Scientific Calculator in C

Console-Based Computational Tool for Academic Excellence

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Project Engine

Exploring the core architecture and logic flow

| Core System Features



Basic Math

Standard arithmetic operations including addition, subtraction, and multi-operand product functions.



Scientific Ops

Advanced calculations using `math.h`: Trigonometry, Logarithms, and Exponents (`pow`, `sqrt`).



Utility Tools

Unit conversions (Temp, Length) and number system transformations (Decimal to Binary).

The Technical Stack

Standard C Framework

The project utilizes standard C libraries to ensure high performance and low-level control over memory and operations.



C

Chosen for its robustness in

Language: numerical computing.



math.h: Leveraged for precise transcendental functions.



stdio.h: Handling structured I/O for the console UI.



C Code Syntax

| System Implementation

- ☰ **Menu-Driven Interface:** A dynamic do-while loop ensures the user can perform continuous calculations without restarting the program.
- 🔗 **Switch-Case Logic:** An efficient decision-making structure routes user input to the specific mathematical module required.
- 🛡️ **Input Validation:** Robust error handling for division-by-zero scenarios and invalid menu selections to prevent runtime crashes.
- ✏️ **Modular Functions:** Each operation is encapsulated in its own function, making the code maintainable and easy to debug.

| Operation Complexity Breakdown

Basic Arithmetic

Trig Functions

Unit Conversions

History Logs

Note: Bars represent the estimated percentage of code logic dedicated to each feature group.

Group Responsibilities

Member Name	Primary Responsibility	Specific Task
Safayat Arefeen	Lead Developer	Main Engine & Menu Logic
Sabbir Ahmed	Scientific Logic	Trigonometric & Log Functions
Samin Yasar Ahmed	Utility Integration	Unit Converters & Conversions
Samiul Islam Sami	QA & Debugging	Input Validation & Error Handling

| Project Statistics

20+

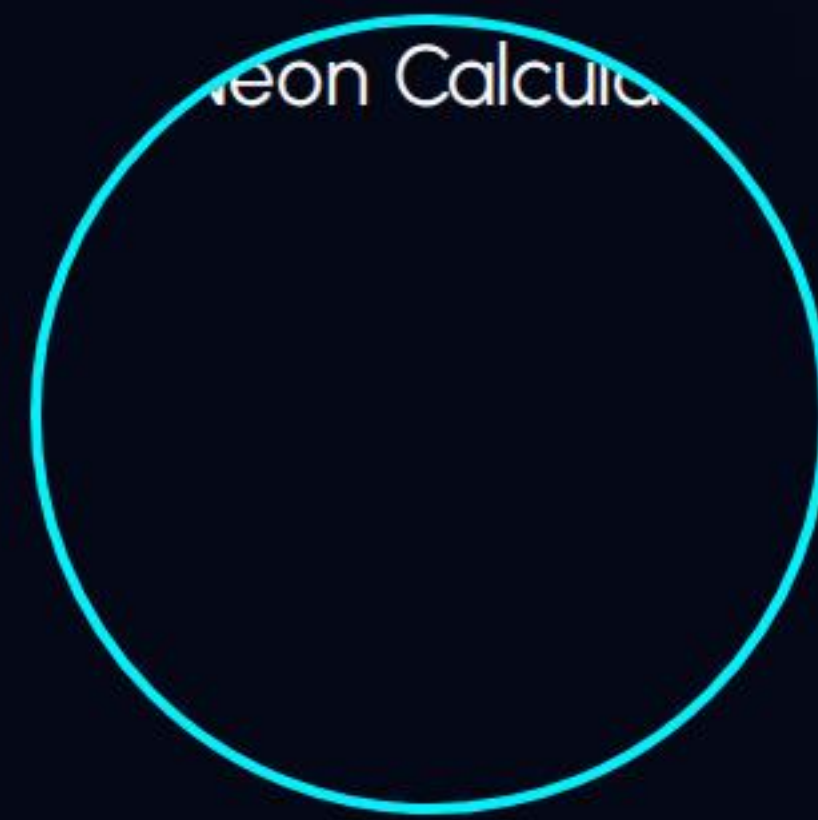
Mathematical Functions

By the Numbers

 500+ Lines of Source Code

 4 Team Members Collaborating

 100% Core Objective Completion



Future Vision

Scaling beyond the console with GUI and File Systems

Q&A

Do you have any questions about our C implementation?



Contact Safayat



Visit NSU CSE

 Floating Math Symbols

Image Sources



<https://jaarchitects.com.bd/wp-content/uploads/2016/12/5-1.jpg>

Source: jaarchitects.com.bd



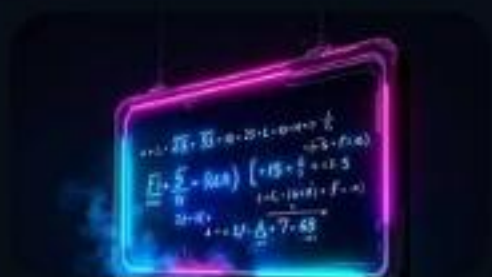
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